

CHAGYL, Turkmenistan

WMO: 385110

Lat: 40.780N

Lon: 55.382E

Elev: 115

StdP: 99.95

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.0	-12.7	-19.9	0.6	-15.2	-16.7	0.9	-11.7	10.0	1.7	9.0	3.2	1.8	90	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	41.6	21.5	40.2	21.4	38.8	20.7	25.0	36.2	23.4	35.4	22.2	34.8	3.4	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.4	16.3	32.4	19.3	14.2	31.1	17.7	12.9	28.7	76.8	35.3	70.2	35.2	65.7	34.7	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
9.6	8.5	7.7	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-18.8	43.9	5.7	1.6	-22.9	45.1	-26.3	46.0	-29.5	46.9	-33.7	48.1		
			-19.1	26.0	5.5	1.8	-23.1	27.2	-26.4	28.3	-29.5	29.2	-33.5	30.5		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.4	0.0	0.8	9.1	15.5	23.6	28.8	31.6	29.3	22.9	14.3	6.3	1.7
	DBStd	12.09	5.65	6.25	4.41	4.61	3.28	3.19	2.86	3.35	3.56	4.80	5.08	4.72
	HDD10.0	1048	310	260	69	8	0	0	0	0	0	17	126	258
	HDD18.3	2478	568	492	286	109	3	0	0	0	4	140	360	516
	CDD10.0	3026	1	2	42	171	422	565	669	599	388	151	16	1
	CDD18.3	1416	0	0	1	23	167	315	411	341	143	16	0	0
	CDH23.3	20027	0	0	28	289	1856	4498	6436	5052	1679	183	4	0
	CDH26.7	11915	0	0	5	101	873	2735	4212	3164	778	46	1	0
Wind	WSAvg	3.2	2.8	3.3	3.4	3.8	3.5	3.5	3.2	3.1	3.4	2.9	2.8	3.1
Precipitation	PrecAvg	150	19	21	28	16	13	5	4	2	4	7	16	15
	PrecMax	218	47	46	61	35	35	19	15	11	23	23	52	38
	PrecMin	91	3	4	10	2	0	0	0	0	0	0	2	1
	PrecStd	38	11	10	13	9	10	5	4	2	6	6	12	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.3	16.7	27.1	33.4	37.4	41.8	43.9	43.1	39.5	31.0	22.1	15.8
		MCWB	9.2	10.7	15.0	18.2	19.8	21.0	22.8	21.2	23.7	16.6	12.6	8.6
	2%	DB	10.8	13.1	22.8	29.4	35.2	40.4	42.2	40.6	35.2	27.6	18.8	12.1
		MCWB	7.2	8.6	13.3	16.7	18.9	20.7	21.8	21.4	18.4	15.7	11.0	8.0
	5%	DB	8.1	10.4	19.8	26.7	33.5	38.7	40.7	39.2	33.2	25.2	16.2	9.7
		MCWB	5.8	6.8	12.2	15.3	18.2	20.3	21.8	20.6	18.2	14.5	10.2	7.0
	10%	DB	6.3	8.0	16.9	23.9	31.5	37.1	39.2	37.7	31.1	22.9	14.0	7.9
		MCWB	4.5	5.4	10.5	13.7	17.2	19.5	21.2	20.1	17.2	13.2	9.1	5.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	11.3	16.0	21.4	21.4	23.6	27.1	26.9	24.9	19.4	15.0	10.1
		MCDB	13.9	16.1	24.4	30.1	33.8	36.4	36.9	38.4	35.2	27.4	19.6	13.8
	2%	WB	7.4	9.0	13.8	18.1	20.3	22.5	25.7	24.3	21.0	17.0	12.3	8.5
		MCDB	10.1	12.8	21.6	26.6	31.5	35.7	37.0	35.5	31.9	25.4	16.9	11.4
	5%	WB	5.9	7.1	12.4	16.1	19.4	21.6	24.0	22.8	19.4	15.5	10.7	7.2
		MCDB	7.9	9.8	19.3	24.0	29.8	34.8	35.3	35.3	29.8	22.6	14.8	9.6
	10%	WB	4.5	5.7	10.9	14.8	18.5	20.7	22.7	21.4	18.0	14.2	9.3	5.8
		MCDB	6.2	7.8	16.4	22.7	28.2	33.7	35.3	35.1	29.2	21.4	13.3	7.8
Mean Daily Temperature Range	5% DB	MDBR	8.0	8.8	11.4	11.7	12.5	13.2	12.8	13.8	13.9	12.8	10.5	8.0
		MCDBR	11.0	12.6	16.6	16.5	15.7	15.5	15.1	16.0	16.8	15.9	14.6	11.4
		MCWBR	8.0	8.4	8.8	7.8	6.6	6.4	6.3	7.0	8.1	8.5	8.9	8.1
	5% WB	MCDBR	10.8	11.8	15.6	14.6	13.5	13.7	12.8	14.1	15.1	13.6	12.8	10.4
		MCWBR	8.0	8.2	8.5	7.7	6.2	6.3	6.4	7.6	8.5	8.2	8.7	7.7
Clear-Sky Solar Irradiance	taub	0.400	0.439	0.492	0.564	0.548	0.540	0.549	0.513	0.485	0.472	0.424	0.398	
	taud	2.027	1.928	1.839	1.716	1.792	1.820	1.809	1.875	1.906	1.916	2.008	2.032	
	Ebn at Noon	704	737	746	723	750	757	744	757	744	692	669	666	
	Edh at Noon	120	153	187	227	215	209	210	191	173	153	121	111	
All-Sky Solar Radiation	RadAvg	1.78	2.67	4.10	5.25	6.58	7.54	7.16	6.59	5.41	3.65	2.37	1.60	
	RadStd	0.23	0.37	0.32	0.51	0.31	0.24	0.27	0.29	0.20	0.26	0.24	0.22	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page